

Parth Mital

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🐙 github.com/parthmital 🌐 parthmital-portfolio.vercel.app

Education & Certifications

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

OCI 2025 Generative AI Professional

Oracle Certified Professional; ID: 103485063OCI25GAIOCP

July 2026

Valid until July 2028

Experience

Concept Marketing

June 2026 – July 2026

Sole Software Developer & Maintainer (Contract)

Vadodara, India

- Migrated legacy on-premise ASP.NET ERP to 24/7 Next.js, TypeScript, Prisma, & PostgreSQL on Vercel + Supabase; **ported 114K+ records across 42 tables with zero data loss**, eliminating single-PC server bottlenecks.
- Partnered directly with operational staff (6–8 hrs daily software usage) to **consolidate fragmented tools into unified workflows** for technician dispatch, AMC contracts, billing, inventory variances, & automated PDF invoicing.
- Secured internet-facing operations with role-based route access, CSRF validation, & PostgreSQL rate limits; scheduled **AES-256 encrypted offsite backups to Google Drive** with 30-day retention via GitHub Actions.
- Engineered test suite with **587 Vitest unit & route tests across 85 files**; delivered production platform **managing 2.8K clients, 28K service calls, & 60+ users** with ongoing solo maintenance.

LearnLine Edustation (IntelGrader)

June 2026 – July 2026

AI Engineer Intern

Remote

- Architected native Android capture app in Kotlin, Jetpack Compose, CameraX, & OpenCV; implemented **real-time quad stability auto-capture across 7 stable frames**, perspective rectification, & native JavaScript bridge.
- Trained multi-task MobileNetV3 to score image quality & 7 defect flags; applied float16 TFLite quantization, **shrinking model size by 82.5% to 2.19 MB** with **2.55 ms edge inference latency**.
- Fine-tuned YOLOv8s struck-out text detector via 2-phase training; integrated downstream masking to **cut OCR token clutter by 7.11%** while maintaining **80% to 96% semantic similarity via RapidFuzz**.
- Designed native PyMuPDF pipeline **extracting 1,033 questions & 383 visuals**; generated dual-GPU EasyOCR cache across **4,574 textbook pages** with AWS Bedrock structured LLM parsing.
- Directed technical implementation with co-founders to deploy **autonomous educational web generator across 6 archetypes**, integrating AWS Bedrock LessonSpec synthesis with Playwright & **axe-core accessibility gates**.

Projects & Research

🔗 NeuroScan AI: Brain Tumour Benchmark & Clinical Platform

PyTorch, FastAPI, React, VTK.js

- Benchmarked **21 deep learning architectures** across 3 stages in a 2-person team under faculty supervision; co-authored **2 research manuscripts (Elsevier format)** currently under review.
- Trained Stage 1 binary detector (**99.59% accuracy**) & Stage 2 subtyper (**98.58% accuracy**) across 4,705 scans, **eliminating 19.3% cross-dataset leakage** via pHash deduplication.
- Evaluated Stage 3 3D volumetric segmentation on 368 BraTS cases; deployed VNet reaching **0.7655 mean Dice (0.8541 WT)** with only **1.4M parameters & 16.9 min training**.
- Developed full-stack clinical platform for **interactive multi-planar 3D slice navigation & mask overlays**; integrated NiBabel NIfTI parsing, VTK.js rendering, & **automated PDF diagnostic reports**.

Technical Skills

Languages: Python, Java, C, C++, SQL, Bash, JavaScript

AI, ML & Data Science: Scikit-learn, ONNX Runtime, Hugging Face, Sentence-Transformers, MONAI, timm, Pandas, NumPy

Frameworks & Web Systems: Node.js, Tailwind CSS, shadcn/ui, TanStack Query, SQLAlchemy, SQLite, RESTful APIs

DevOps, Cloud & Tooling: Docker, Git, GitHub Actions, Linux, CI/CD, Postman, Vite, Agile

Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering, Compiler Design, Cloud Architecture, Cryptography & Network Security, Natural Language Processing, Digital Image Processing, Linear Algebra, Probability & Statistics